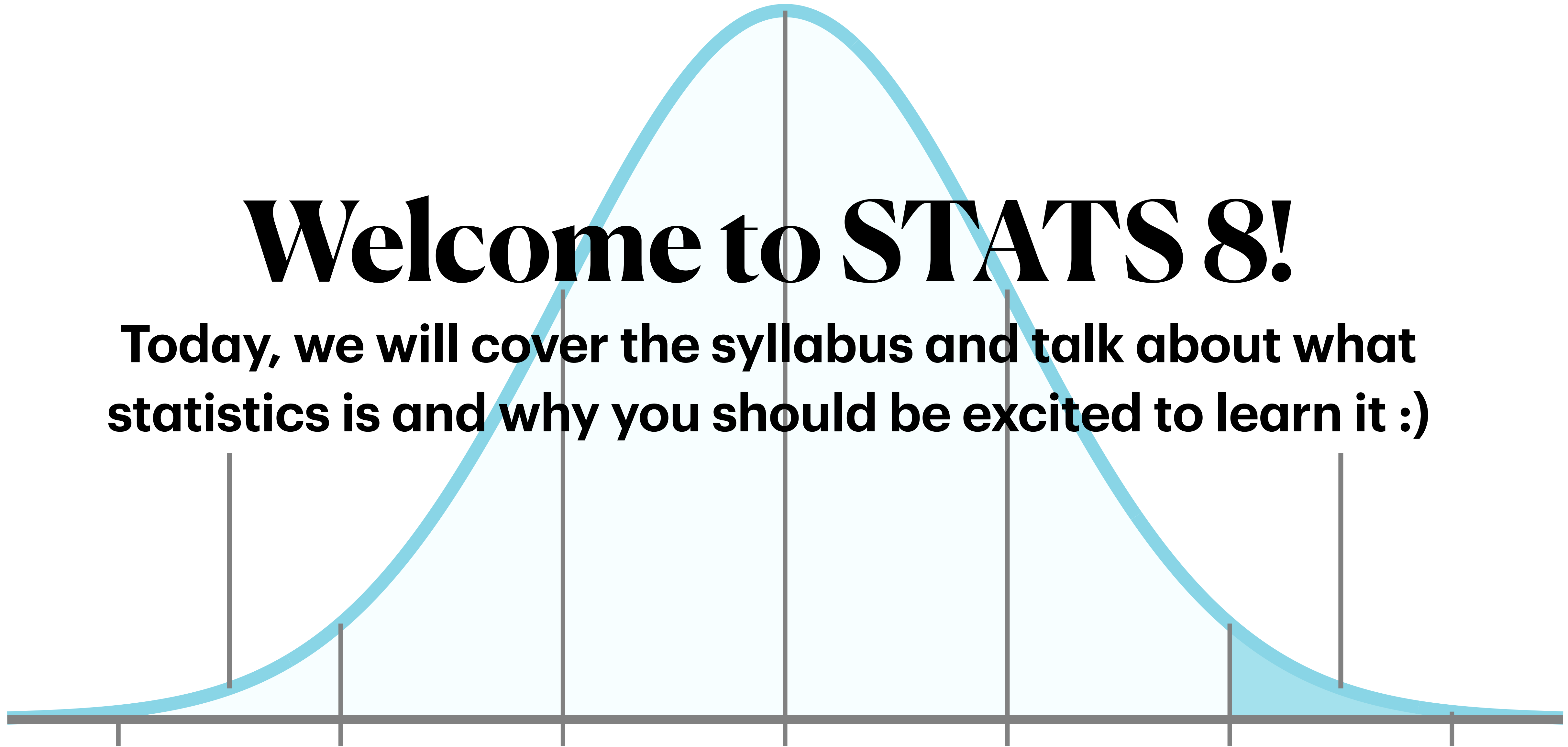


Welcome to STATS 8!

Today, we will cover the syllabus and talk about what statistics is and why you should be excited to learn it :)



Who are we?

- Instructor: Jessalyn (me), she/her/hers
- TAs:
 - Asha Pantula
 - Alexandria Richardson
 - Xiaoke Cao
 - Noah Benjamin (grader)
- All of us are PhD students in Statistics here

UCI
Statistics

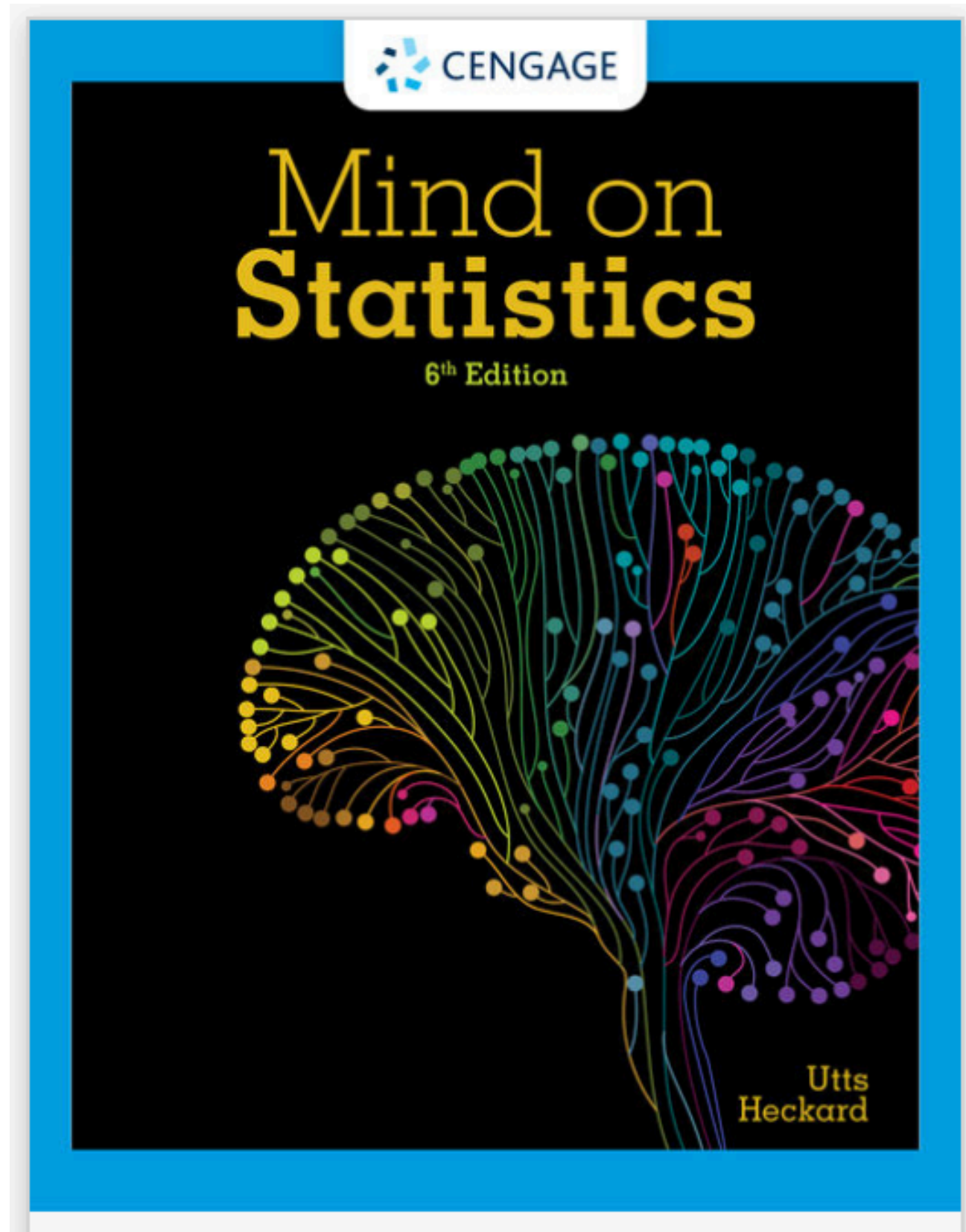


Syllabus (Policies)

First Assignment

- There is a Google form linked on the Canvas page at the beginning of the course
 - What is your math/stat background, if any?
 - What sorts of things are you interested in?
 - What are you worried about with this class?
 - What kind of extra support might you need?
 - etc.

Required Materials



- *Mind On Statistics*, 6th edition with WebAssign by Jessica Utts and Robert Heckard
 - You need WebAssign access with it for the homework!
 - You can get a 2-week trial of this for free if you're not sure you'll stay in the class
 - For material — trust my lecture slides over the book (there will be some minor differences, and I will point them out as we go)
- Scientific calculator (not a graphing calculator) for exams

Technology

- We will be using the R programming language for computation
 - (It will not be more complicated than using a graphing calculator)
- We will be using PollEverywhere in lectures, so you need to bring a device that can connect to the internet
 - This will be used to take attendance (not for today, for future lectures)
 - **Let's practice this real quick:** [PollEv.com/stats8](https://pollev.com/stats8)



Poll Everywhere

Grade Breakdown

- Exams (80%)
 - Two midterms, each 20%
 - Final exam, 40%
 - If your final score is better than at least one of your midterm scores, it may replace the lower score
- Homework (via WebAssign), 10%
- Discussion section assignments, 10%
- Lecture attendance, <0.5%
 - *If* you are at most 0.5 percentage points away from the next letter grade (including +/-) *and* you attended at least 23 lectures your grade will automatically be rounded up
 - **No other grade rounding requests will be considered.**

A Couple Notes on Exams

- They are not only *assessment*, they are intended to be *motivation* to try to understand the material
 - If there were only take-home exams or projects, wouldn't you be tempted to not learn the material and instead use ChatGPT/Gemini/Claude to finish your work? (I would! I'm busy!)
- I don't like hard exams and I don't like curving grades; the goal is to write exams that test expected skills
- Where the book differs from my slides, go with my slides

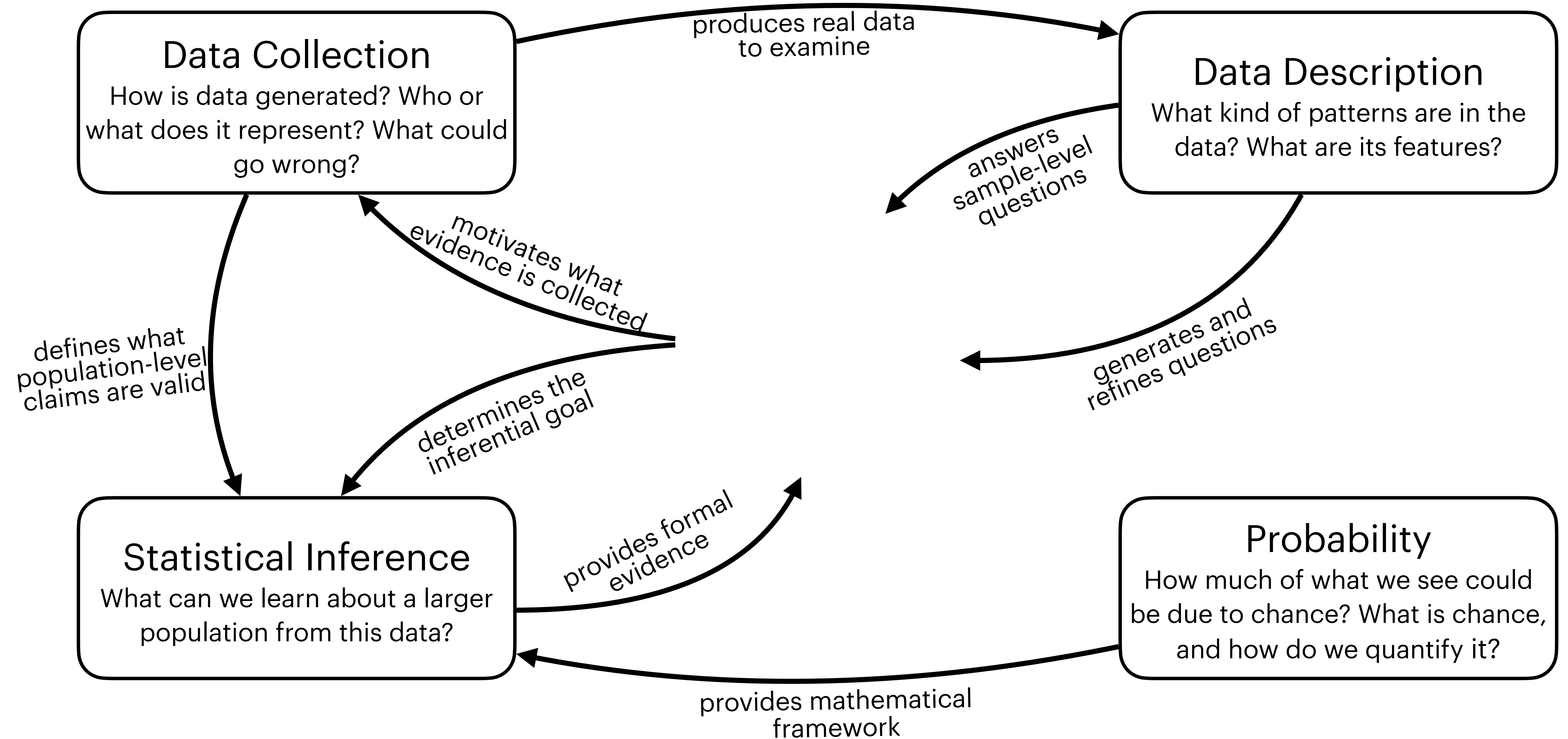
Discussion Sections

- Small group activities (3-5 students each)
 - Not required to stay the same throughout the quarter
 - (But may be easiest)
 - On Wednesday, you will be asked to talk to people near you, so mentally prepare yourself
- Attendance is required, and your work will be submitted for credit

Class Structure

Friday, Week X	Monday, Week X+1	Wednesday, Week X+1	Friday, Week X+1
Lecture at 11a Homework due at 11:59p, on topics from Week X-1 Start suggested reading on topics for week X+1 (not required, but can be useful)	Lecture at 11a	Lecture at 11a Discussion at 9a, 10a, 1p, or 4p; will be an activity using topics from Week X	Lecture at 11a Homework due at 11:59p, on topics from Week X Start suggested reading on topics for Week X+2

What is statistics?

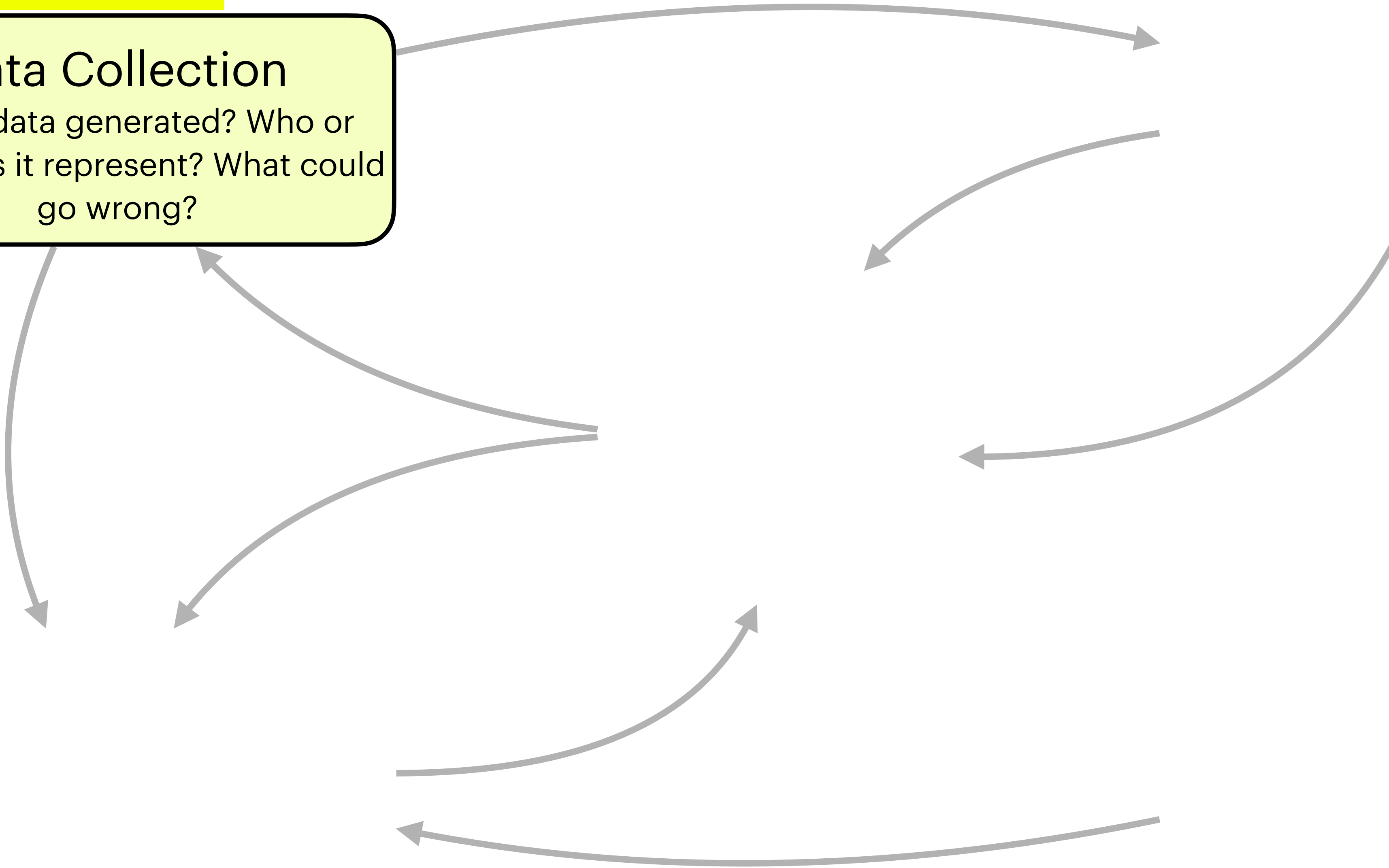


Syllabus (Topics)

Week 1 - 2

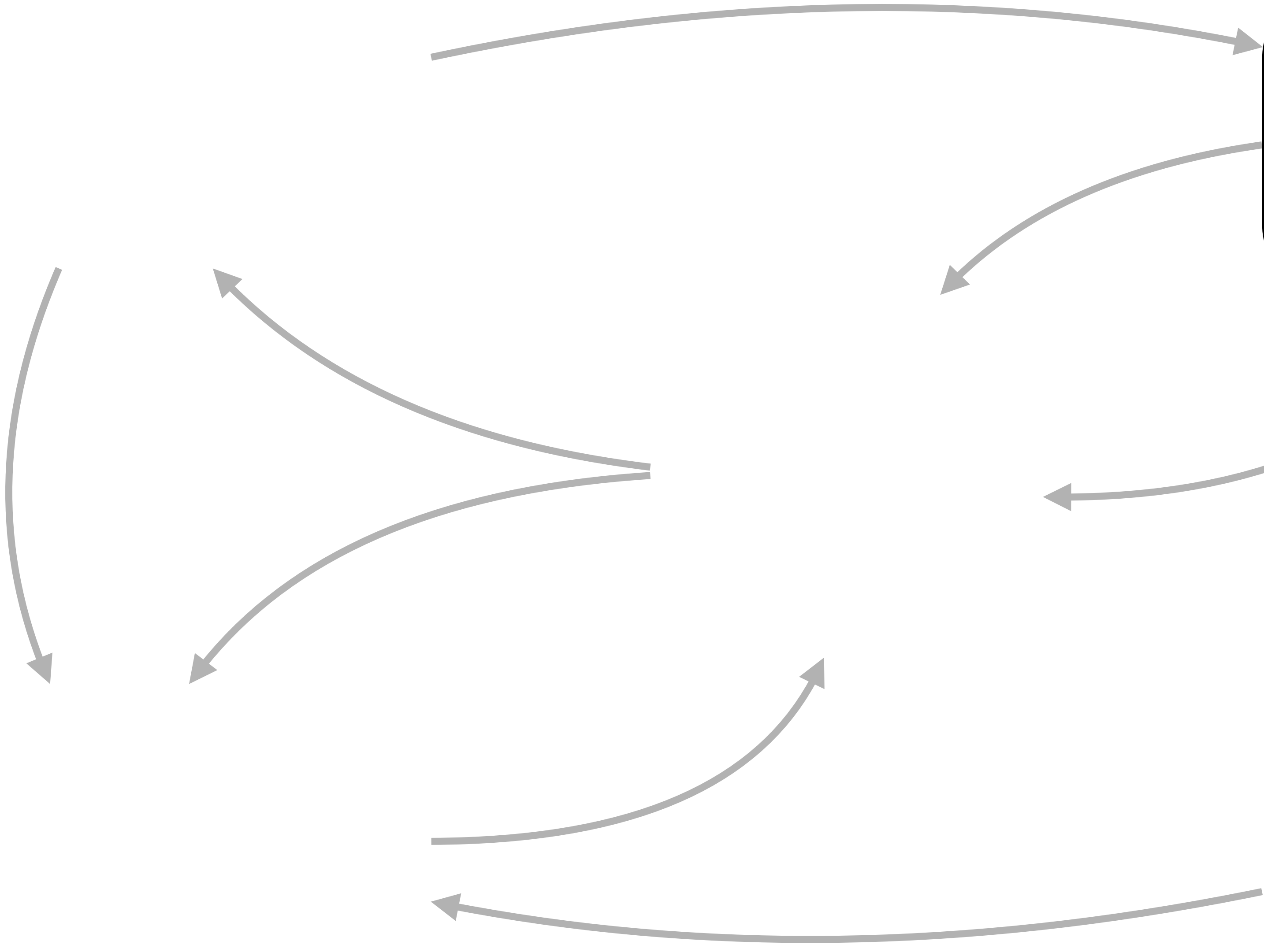
Data Collection

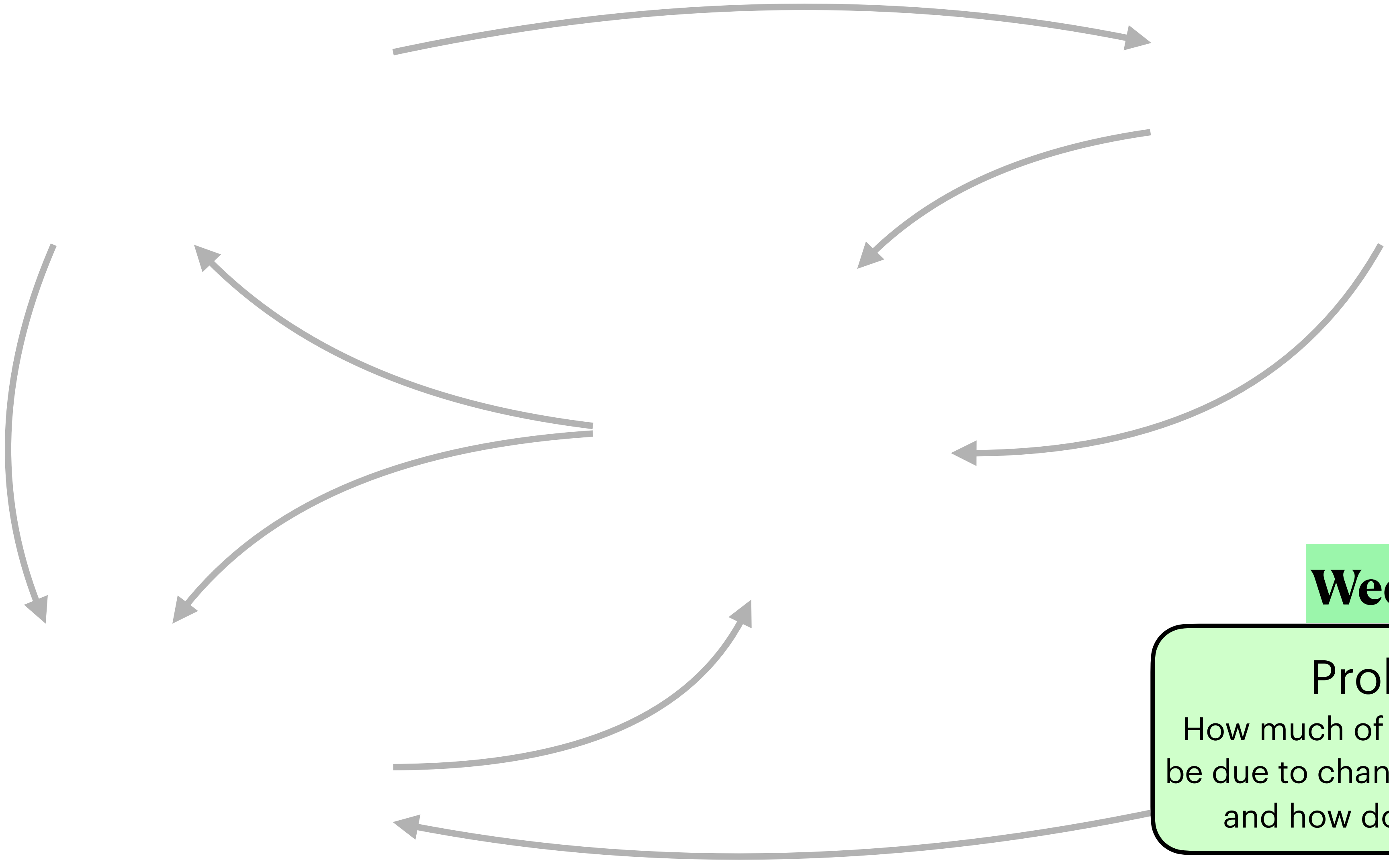
How is data generated? Who or what does it represent? What could go wrong?



Week 2 - 3

Data Description
What kind of patterns are in the data? What are its features?





Week 4 - 6

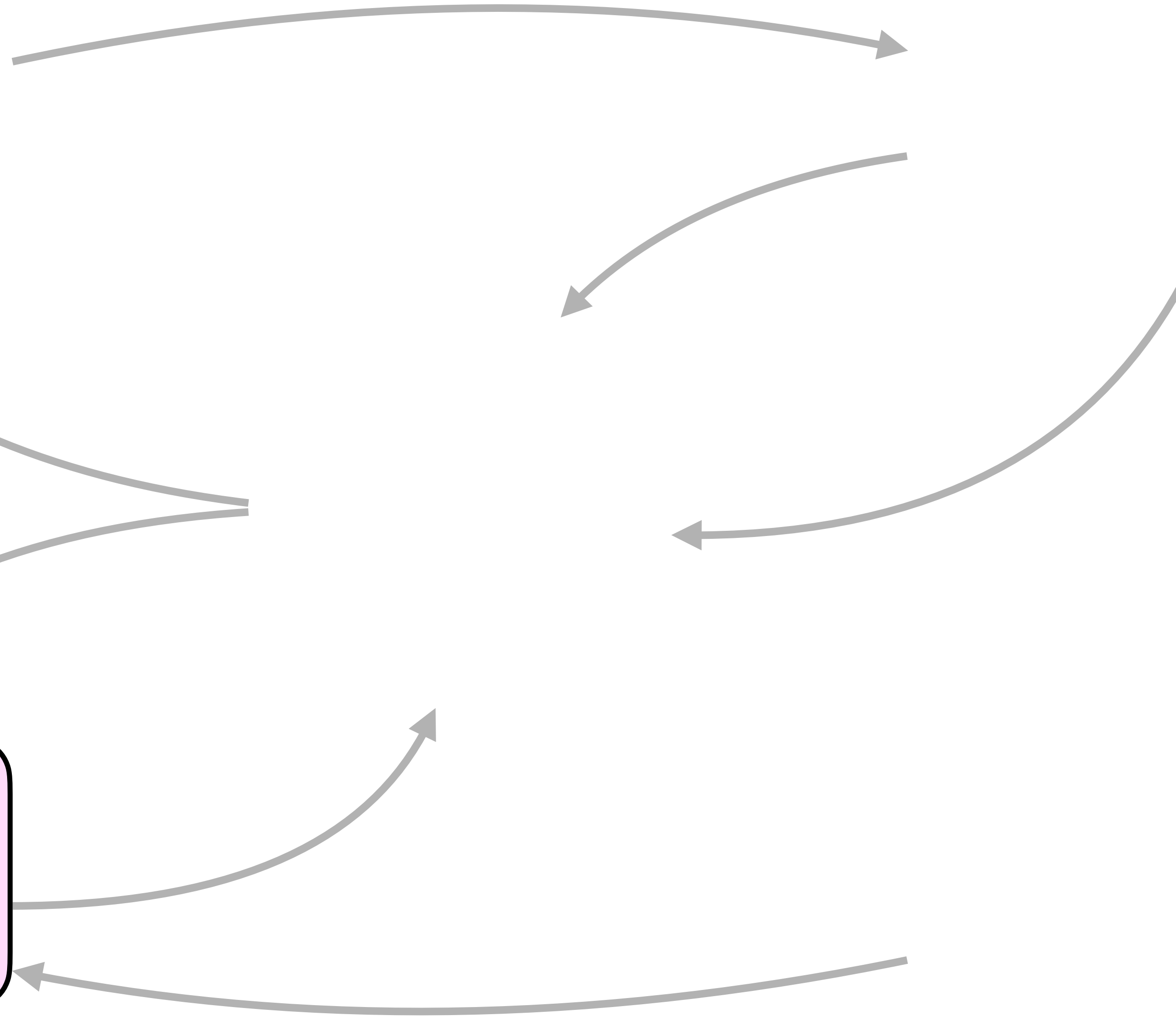
Probability

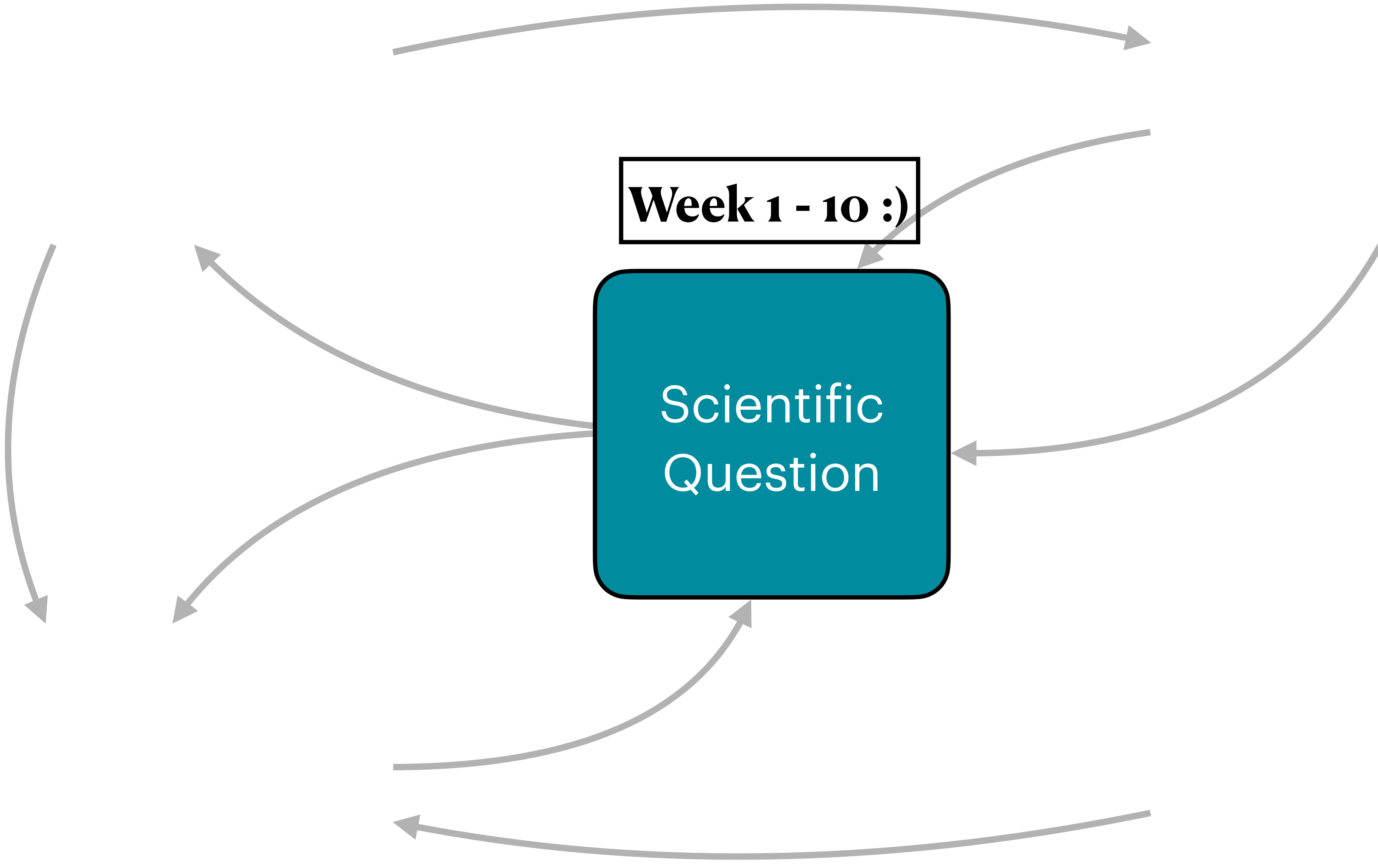
How much of what we see could be due to chance? What is chance, and how do we quantify it?

Week 7 - 10

Statistical Inference

What can we learn about a larger population from this data?





My goals for this class

- By the time you leave STATS 8...
 - You *absolutely, positively need* to..
 - Be proficient at interpreting the results of statistical analyses
 - Reading difficult plots, understanding why studies are designed the way they are and what can go wrong, interpreting probabilities and numerical results of inference
- I would also *prefer* for you to be able to...
 - Be able to do basic statistical computations, preferably in R
 - Be able to explore data using numeric summaries, plots, specialized vocabulary
 - Be able to perform statistical inference procedures covered in this class

Welcome to the Class!

My opinion: this class can change the way that you see the world, **if you're willing to let it!**



Next: data collection, study design